

AZURE LOAD BALANCER VS APPLICATION GATEWAY

Design Comparison and Cloud Implementation

PROJECT DOCUMENTATION

Azure Cloud Networking Project

1. Abstract

Cloud computing has become an important technology for developing and deploying modern applications. As the number of users accessing cloud applications increases, applications need to handle large amounts of network traffic without affecting performance, availability, or reliability.

Load balancing is an important technique used in cloud environments to distribute incoming traffic across multiple backend servers or resources. Microsoft Azure provides different services for managing and distributing network traffic, among which Azure Load Balancer and Azure Application Gateway are widely used.

Azure Load Balancer is a Layer 4 load-balancing service that distributes traffic based on information such as IP addresses and ports. It supports TCP and UDP traffic and can distribute traffic across multiple virtual machines and virtual machine scale sets using health probes.

Azure Application Gateway is a Layer 7 web traffic load-balancing service designed for HTTP and HTTPS applications. It can make routing decisions using URL paths and host names. It also supports SSL/TLS termination and can work with Web Application Firewall capabilities.

This project compares both services in terms of networking layer, traffic type, routing, security features, availability, and use cases.

2. Introduction

In today's cloud computing environment, applications are expected to handle a large number of users, requests, and data transactions while providing high availability and reliable performance.

Azure Load Balancer works primarily at Layer 4 and distributes network traffic using information such as IP addresses and ports. It supports TCP and UDP and is designed for high availability and efficient network-level traffic distribution.

Azure Application Gateway works at Layer 7, the application layer. It understands web requests and can make routing decisions based on URL paths, host names, and other HTTP-related information.

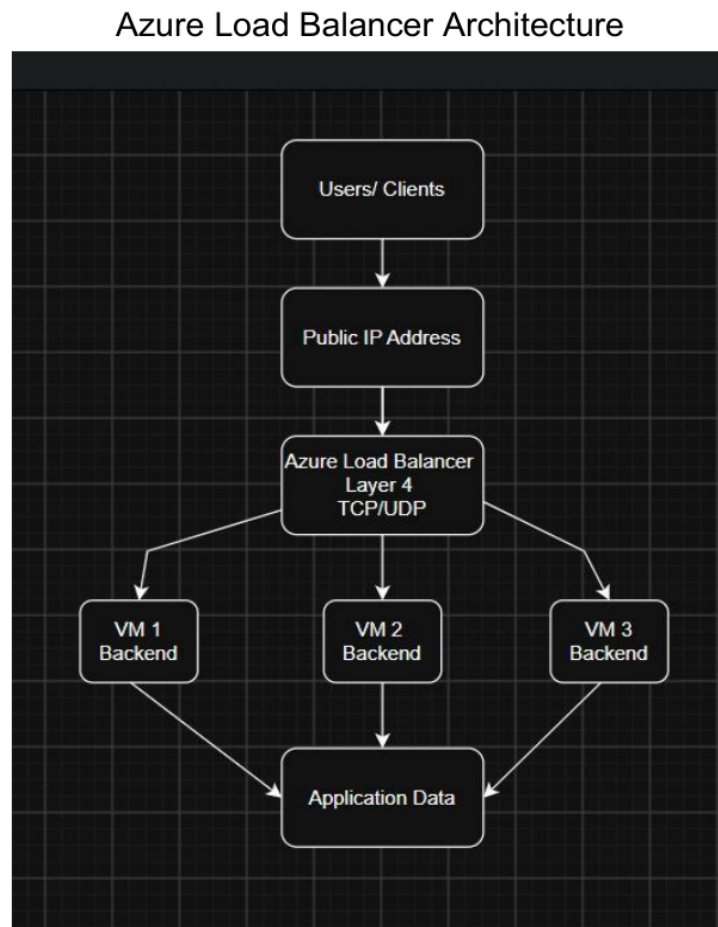
Application Gateway also supports SSL/TLS termination and can integrate with Web Application Firewall capabilities. In modern cloud architectures, both services may be used together to handle different networking requirements.

The project focuses on understanding the architecture, Azure services, implementation tasks, modules, and differences between these two load-balancing solutions.

3. Architecture

The project contains two architecture designs: Azure Load Balancer and Azure Application Gateway.

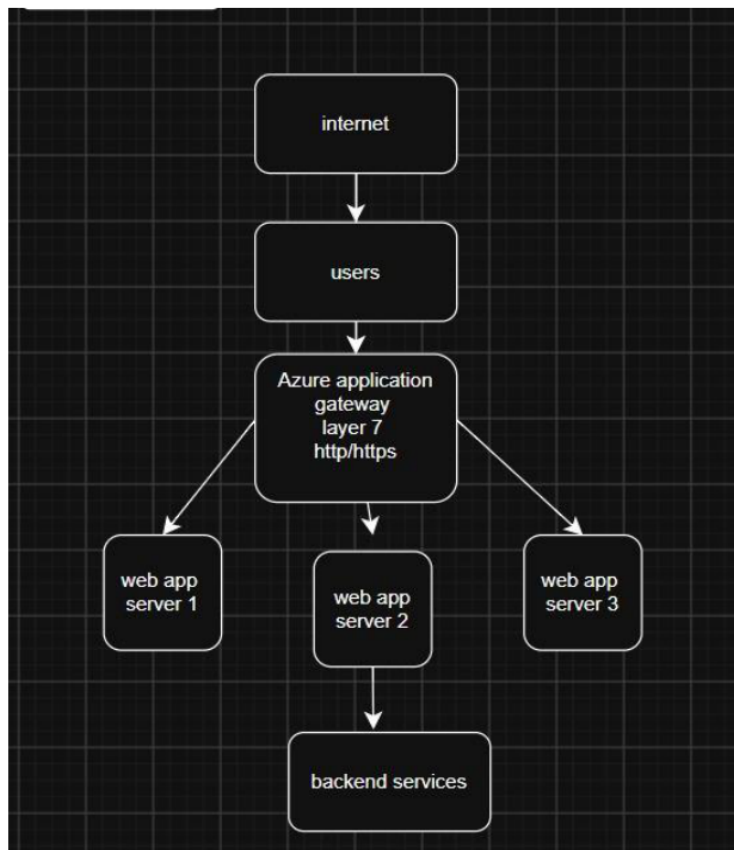
3.1 Azure Load Balancer Architecture



Explanation: Client traffic reaches the Public IP Address and is distributed by the Azure Load Balancer to backend virtual machines. Health probes help identify healthy backend instances.

3.2 Azure Application Gateway Architecture

Application gateway



Explanation: Internet users access the application through Application Gateway. The gateway handles HTTP/HTTPS traffic and forwards requests to the appropriate backend web application servers.

4. Azure Services Required to Implement the Project

The following Azure services and components are required or useful for implementing and testing the proposed architecture.

1. Azure Load Balancer

Layer 4 traffic distribution for TCP/UDP traffic across backend resources.

2. Azure Application Gateway

Layer 7 HTTP/HTTPS load balancing with application-aware routing and SSL/TLS termination.

3. Azure Virtual Network (VNet)

Provides the private networking environment for the Azure resources.

4. Azure Virtual Machines / VM Scale Sets

Provide multiple backend instances for load-balancing and availability testing.

5. Public IP Address

Provides an internet-facing endpoint for the public load-balancing architecture.

6. Health Probes

Check backend availability and help direct traffic to healthy resources.

7. Network Security Group (NSG)

Controls allowed inbound and outbound network traffic.

8. Web Application Firewall (WAF)

Can be integrated with Application Gateway to inspect and protect web traffic.

9. Azure Monitor

Helps observe resource health, traffic behavior, and performance during testing.

Note: The exact services used can depend on the implementation scope and Azure environment.

5. Design Comparison

Feature	Azure Load Balancer	Azure Application Gateway
Networking Layer	Layer 4	Layer 7
Main Traffic	TCP / UDP	HTTP / HTTPS
Routing Basis	IP address and port	URL path, host name and HTTP information
Application Awareness	Network-level	Application-aware
SSL/TLS Termination	Not the main feature	Supported
WAF	Not the primary feature	Can integrate with WAF
Main Purpose	Network-level traffic distribution	Web traffic management and routing

6. FIVE TASKS

Task 1 – Azure Network Setup

Create the Resource Group, Virtual Network, subnets, and required network security rules.

Task 2 – Load Balancer Implementation

Create the public Load Balancer, frontend public IP, backend pool, load-balancing rule, and health probe.

Task 3 – Application Gateway Implementation

Create Application Gateway with frontend access, backend pool, HTTP/HTTPS listener, routing rules, and health checks.

Task 4 – Traffic Routing and Testing

Deploy multiple backend servers and test how requests are distributed through both services.

Task 5 – Comparison and Documentation

Compare Layer 4 and Layer 7 behavior, document observations, and prepare the final PPT and report.

7. Project Modules

Module 1 – Network Configuration

Resource group, VNet, subnets, Public IP, and NSG.

Module 2 – Backend Server Module

Creation and configuration of multiple backend instances.

Module 3 – Azure Load Balancer Module

Frontend IP, backend pool, health probe, and Layer 4 rule.

Module 4 – Application Gateway Module

Frontend IP, listener, backend pool, and Layer 7 routing.

Module 5 – Monitoring and Testing

Traffic tests, backend health verification, and availability checks.

Module 6 – Documentation and Presentation

Architecture explanation, report, Git repository, and PPT.

8. Conclusion

Azure Load Balancer and Azure Application Gateway provide different approaches to traffic distribution. Azure Load Balancer focuses on high-performance Layer 4 network traffic distribution, while Application Gateway provides Layer 7 web traffic management and intelligent routing. Understanding their architecture and capabilities helps in designing scalable and highly available Azure applications.

9. Final Submission Checklist

- GitHub project link
- Project Abstract
- Project Introduction
- Architecture Diagram with Explanation
- Azure Services Required to Implement the Project
- Five Tasks
- Project Modules
- Project PPT / Presentation
- Final Project Report